

Concise Study Guide: Number Theory (Chapter 4, Sections 4.1-4.4)

Based on slides for Chapter 4

Section 4.1: Divisibility and Modular Arithmetic

Division

Theorem/Definition

Definition: If a and b are integers with $a \neq 0$, then a **divides** b if there exists an integer c such that $b = ac$.

- Notation: $a \mid b$ (a divides b), $a \nmid b$ (a does not divide b).
- Terminology: If $a \mid b$, then a is a **factor** or **divisor** of b , and b is a **multiple** of a .

Question/Example

Determine whether $3 \mid 7$ and whether $3 \mid 12$. **Solution:** $3 \nmid 7$ because $7/3$ is not an integer. $3 \mid 12$ because $12 = 3 \cdot 4$.

Theorem/Definition

Theorem 1 (Properties of Divisibility): Let a, b, c be integers with $a \neq 0$.

1. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
2. If $a \mid b$, then $a \mid bc$ for all integers c .
3. If $a \mid b$ and $b \mid c$, then $a \mid c$ (Transitivity).

Corollary: If $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for any integers m, n .

The Division Algorithm

Theorem/Definition

Theorem (Division Algorithm): If a is an integer and d is a positive integer, then there exist unique integers q (quotient) and r (remainder) such that $a = dq + r$ and $0 \leq r < d$.

- d : divisor
- a : dividend
- $q = a \operatorname{div} d$
- $r = a \operatorname{mod} d$

Question/Example

What are the quotient and remainder when:

1. 101 is divided by 11? **Solution:** $101 = 11 \cdot 9 + 2$. So, $q = 9$ and $r = 2$. ($101 \text{ div } 11 = 9$, $101 \bmod 11 = 2$).
2. -11 is divided by 3? **Solution:** $-11 = 3 \cdot (-4) + 1$. So, $q = -4$ and $r = 1$. ($-11 \text{ div } 3 = -4$, $-11 \bmod 3 = 1$). Note that the remainder r must be non-negative ($0 \leq r < d$).

Modular Arithmetic

Theorem/Definition

Definition (Congruence): If a and b are integers and m is a positive integer, then a is **congruent to b modulo m** if $m \mid (a - b)$.

- Notation: $a \equiv b \pmod{m}$.
- m is the **modulus**.
- If not congruent: $a \not\equiv b \pmod{m}$.
- $a \equiv b \pmod{m}$ if and only if a and b have the same remainder when divided by m .

Question/Example

1. Is $17 \equiv 5 \pmod{6}$? **Solution:** Yes, because $6 \mid (17 - 5)$, since $6 \mid 12$.
2. Is $24 \equiv 14 \pmod{6}$? **Solution:** No, because $6 \nmid (24 - 14)$, since $6 \nmid 10$.

Theorem/Definition

Theorem 4: $a \equiv b \pmod{m}$ if and only if $a = b + km$ for some integer k .

Theorem 3: $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$. (Note the difference: $\pmod{m}$ indicates a relation, $\bmod m$ indicates the function giving the remainder).

Theorem/Definition

Theorem 5 (Congruences of Sums and Products): Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

- $a + c \equiv b + d \pmod{m}$
- $ac \equiv bd \pmod{m}$

Question/Example

Given $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$. Verify Theorem 5. **Solution:**

- Sum: $7 + 11 = 18$. $2 + 1 = 3$. Is $18 \equiv 3 \pmod{5}$? Yes, since $5 \mid (18 - 3) = 15$.
- Product: $7 \cdot 11 = 77$. $2 \cdot 1 = 2$. Is $77 \equiv 2 \pmod{5}$? Yes, since $5 \mid (77 - 2) = 75$.

Theorem/Definition

Corollary (Computing mod m of Sums/Products):

- $(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$
- $(a \cdot b) \bmod m = ((a \bmod m) \cdot (b \bmod m)) \bmod m$

Arithmetic Modulo m

Theorem/Definition

Definitions: Let $Z_m = \{0, 1, \dots, m-1\}$.

- Addition modulo m : $a +_m b = (a + b) \bmod m$.
- Multiplication modulo m : $a \cdot_m b = (a \cdot b) \bmod m$.

Properties: These operations satisfy closure, associativity, commutativity, identity (0 for $+$, 1 for $\cdot$), additive inverses ($m - a$ is inverse of $a \neq 0$), and distributivity within Z_m . (Multiplicative inverses do not always exist).

Question/Example

Find $7 +_{11} 9$ and $7 \cdot_{11} 9$. **Solution:**

- $7 +_{11} 9 = (7 + 9) \bmod 11 = 16 \bmod 11 = 5$.
- $7 \cdot_{11} 9 = (7 \cdot 9) \bmod 11 = 63 \bmod 11 = 8$. (Since $63 = 5 \cdot 11 + 8$).

Section 4.2: Integer Representations and Algorithms

Base b Expansions

Theorem/Definition

Theorem 1: Any positive integer n can be uniquely expressed in base b ($b > 1$) as: $n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b^1 + a_0 b^0$ where $k \geq 0$, $0 \leq a_j < b$ for all j , and $a_k \neq 0$. Notation: $(a_k a_{k-1} \dots a_1 a_0)_b$.

Question/Example

Binary (Base 2): Digits $\{0, 1\}$.

- What is $(101011111)_2$ in decimal? **Solution:** $1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 256 + 64 + 16 + 8 + 4 + 2 + 1 = 351$.
- What is $(11011)_2$ in decimal? **Solution:** $1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 16 + 8 + 0 + 2 + 1 = 27$.

Octal (Base 8): Digits $\{0, \dots, 7\}$.

- What is $(7016)_8$ in decimal? **Solution:** $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 7 \cdot 512 + 0 + 8 + 6 = 3584 + 8 + 6 = 3598$.
- What is $(111)_8$ in decimal? **Solution:** $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$.

Hexadecimal (Base 16): Digits $\{0, \dots, 9, A=10, B=11, C=12, D=13, E=14, F=15\}$.

- What is $(2AE0B)_{16}$ in decimal? **Solution:** $2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 2 \cdot 65536 + 10 \cdot 4096 + 14 \cdot 256 + 0 + 11 = 131072 + 40960 + 3584 + 11 = 175627$.
- What is $(E5)_{16}$ in decimal? **Solution:** $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$.

Base Conversion Algorithm

To convert decimal integer n to base b :

1. Divide n by b . The remainder a_0 is the last digit. Let the quotient be q_0 .
2. Divide q_0 by b . The remainder a_1 is the second-to-last digit. Let the quotient be q_1 .
3. Repeat until the quotient is 0. The remainders, read in reverse order of calculation, form the base b digits.

Pseudocode

```
procedure base_b_expansion( $n, b$ : positive integers with  $b > 1$ )  
   $q := n$   $k := 0$  while ( $q \neq 0$ )  
     $a_k := q \bmod b$   $q := q \div b$   $k := k + 1$  return  $(a_{k-1}, \dots, a_1, a_0)$  {The base  $b$  digits}
```

Question/Example

Find the octal expansion of $(12345)_{10}$. **Solution:**

- $12345 = 8 \cdot 1543 + 1$ ($a_0 = 1$)
- $1543 = 8 \cdot 192 + 7$ ($a_1 = 7$)
- $192 = 8 \cdot 24 + 0$ ($a_2 = 0$)
- $24 = 8 \cdot 3 + 0$ ($a_3 = 0$)
- $3 = 8 \cdot 0 + 3$ ($a_4 = 3$)

Reading remainders from bottom up: $(30071)_8$.

Conversion Between Binary, Octal, Hexadecimal

- Binary $\leftrightarrow$ Octal: Group binary digits in blocks of 3 (since $8 = 2^3$).
- Binary $\leftrightarrow$ Hexadecimal: Group binary digits in blocks of 4 (since $16 = 2^4$).

Question/Example

Find the octal and hexadecimal expansions of $(11111010111100)_2$. **Solution:**

- **Octal:** Group by 3 (add leading 0s): $(011\ 111\ 010\ 111\ 100)_2 = (37274)_8$.
- **Hexadecimal:** Group by 4 (add leading 0s): $(0011\ 1110\ 1011\ 1100)_2 = (3EBC)_{16}$.

Algorithms for Integer Operations

Pseudocode

[title=Binary Addition] **procedure** add(a, b : positive integers with binary expansions $(a_{n-1} \dots a_0)_2, (b_{n-1} \dots b_0)_2$) $c := 0$ {carry} **for** $j := 0$ **to** $n - 1$ $d := \lfloor (a_j + b_j + c)/2 \rfloor$ {new carry} $s_j := a_j + b_j + c - 2d$ {sum bit} $c := d$ $s_n := c$ {final carry} **return** $(s_n s_{n-1} \dots s_0)_2$ {the sum}

Pseudocode

[title=Binary Multiplication (Simplified)] **procedure** multiply(a, b : positive integers with binary expansions $(a_{n-1} \dots a_0)_2, (b_{n-1} \dots b_0)_2$) **for** $j := 0$ **to** $n - 1$ **if** $b_j = 1$ **then** $c_j := a$ shifted j places left **else** $c_j := 0$ $p := 0$ {product} **for** $j := 0$ **to** $n - 1$ $p := p + c_j$ {add partial products} **return** p

Modular Exponentiation

Goal: Compute $b^n \pmod{m}$ efficiently for large n . Method: Use binary expansion of $n = (a_{k-1} \dots a_0)_2$. $b^n = b^{a_{k-1}2^{k-1} + \dots + a_12^1 + a_02^0} = \prod_{a_i=1} b^{2^i}$. Compute powers $b^{2^0}, b^{2^1}, b^{2^2}, \dots$ by repeated squaring, taking mod m at each step.

Pseudocode

[title=Modular Exponentiation] **procedure** modular_exponentiation(b : integer, $n = (a_{k-1} \dots a_0)_2$, m : positive integer) $x := 1$ **power** $:= b \pmod{m}$ **for** $i := 0$ **to** $k - 1$ **if** $a_i = 1$ **then** $x := (x \cdot \text{power}) \pmod{m}$ **power** $:= (\text{power} \cdot \text{power}) \pmod{m}$ **return** x {result is $b^n \pmod{m}$ }

Question/Example

Compute $3^{11} \pmod{m}$ (let's assume m is large, so we just compute 3^{11}). **Solution:** $n = 11 = (1011)_2$. $k = 4$. $a_0 = 1, a_1 = 1, a_2 = 0, a_3 = 1$.

- $i = 0$: $a_0 = 1$. $x = (1 \cdot 3) \pmod{m} = 3$. **power** $= (3 \cdot 3) \pmod{m} = 9$.
- $i = 1$: $a_1 = 1$. $x = (3 \cdot 9) \pmod{m} = 27$. **power** $= (9 \cdot 9) \pmod{m} = 81$.
- $i = 2$: $a_2 = 0$. $x = 27$. **power** $= (81 \cdot 81) \pmod{m} = 6561$.
- $i = 3$: $a_3 = 1$. $x = (27 \cdot 6561) \pmod{m} = 177147$. **power** $= (6561 \cdot 6561) \pmod{m}$.

Return $x = 177147$. (If m was given, mods would be applied at each step).

Section 4.3: Primes and Greatest Common Divisors

Prime Numbers

Theorem/Definition

Definition: A positive integer $p > 1$ is **prime** if its only positive factors are 1 and p . Otherwise, it is **composite**.

Theorem/Definition

The Fundamental Theorem of Arithmetic: Every integer $n > 1$ can be uniquely written as a product of primes, $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$, where $p_1 < p_2 < \dots < p_k$ are primes and $e_i \geq 1$.

Question/Example

Prime factorizations:

- $100 = 2^2 \cdot 5^2$
- $641 = 641$ (prime)
- $999 = 3^3 \cdot 37$
- $1024 = 2^{10}$

Theorem/Definition

Theorem: If n is composite, then it has a prime factor less than or equal to $\sqrt{n}$. **Use:** To check if n is prime, only need to test for divisibility by primes $p \leq \sqrt{n}$.

Theorem/Definition

Theorem (Euclid): There are infinitely many primes. **Proof Idea:** Assume finite primes $p_1, \dots, p_n$. Consider $Q = (p_1 p_2 \dots p_n) + 1$. Q is not divisible by any p_i . So either Q is prime itself, or it has a prime factor not in the list. Contradiction.

Theorem/Definition

Definition (Mersenne Primes): Primes of the form $2^p - 1$, where p is also prime. E.g., $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$. (Note: $2^{11} - 1 = 2047 = 23 \cdot 89$ is not prime). Large known primes are often Mersenne primes.

Theorem/Definition

Prime Number Theorem (Distribution of Primes): The number of primes less than or equal to x , denoted $\pi(x)$, is asymptotically approximated by $x/\ln x$. That is, $\lim_{x \rightarrow \infty} \frac{\pi(x)}{x/\ln x} = 1$.

Greatest Common Divisors (GCD) and Least Common Multiples (LCM)

Theorem/Definition

Definition (GCD): The greatest common divisor of two integers a, b (not both zero), denoted $\gcd(a, b)$, is the largest integer d such that $d \mid a$ and $d \mid b$. **Definition (Relatively Prime):** a and b are relatively prime if $\gcd(a, b) = 1$. **Definition (Pairwise Relatively Prime):** Integers $a_1, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ for all $i \neq j$.

Question/Example

- $\gcd(24, 36) = 12$.
- $\gcd(17, 22) = 1$. (17 and 22 are relatively prime).
- Are 10, 17, 21 pairwise relatively prime? **Solution:** $\gcd(10, 17) = 1$, $\gcd(10, 21) = 1$, $\gcd(17, 21) = 1$. Yes.
- Are 10, 19, 24 pairwise relatively prime? **Solution:** $\gcd(10, 24) = 2 \neq 1$. No.

Theorem/Definition

GCD using Prime Factorization: If $a = p_1^{a_1} \dots p_n^{a_n}$ and $b = p_1^{b_1} \dots p_n^{b_n}$ (where exponents can be 0), then: $\gcd(a, b) = p_1^{\min(a_1, b_1)} \dots p_n^{\min(a_n, b_n)}$.

Definition (LCM): The least common multiple of positive integers a, b , denoted $\text{lcm}(a, b)$, is the smallest positive integer m such that $a \mid m$ and $b \mid m$.

LCM using Prime Factorization: $\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \dots p_n^{\max(a_n, b_n)}$.

Theorem 5: $a \cdot b = \gcd(a, b) \cdot \text{lcm}(a, b)$.

Question/Example

- $a = 120 = 2^3 \cdot 3^1 \cdot 5^1$. $b = 500 = 2^2 \cdot 5^3 = 2^2 \cdot 3^0 \cdot 5^3$.
- $\gcd(120, 500) = 2^{\min(3, 2)} \cdot 3^{\min(1, 0)} \cdot 5^{\min(1, 3)} = 2^2 \cdot 3^0 \cdot 5^1 = 4 \cdot 1 \cdot 5 = 20$.
- $\text{lcm}(120, 500) = 2^{\max(3, 2)} \cdot 3^{\max(1, 0)} \cdot 5^{\max(1, 3)} = 2^3 \cdot 3^1 \cdot 5^3 = 8 \cdot 3 \cdot 125 = 3000$.
- Check Theorem 5: $120 \cdot 500 = 60000$. $\gcd(120, 500) \cdot \text{lcm}(120, 500) = 20 \cdot 3000 = 60000$. It holds.

The Euclidean Algorithm

An efficient algorithm to compute $\gcd(a, b)$. Based on Lemma 1: If $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

Pseudocode

```
[title=Euclidean Algorithm] procedure gcd( $a, b$ : positive integers)  $x := a$   $y := b$  while  $y \neq 0$   
 $r := x \bmod y$   $x := y$   $y := r$  return  $x$  {gcd is the last non-zero remainder}
```

Question/Example

Find $\gcd(287, 91)$. **Solution:**

- $287 = 91 \cdot 3 + 14$
- $91 = 14 \cdot 6 + 7$
- $14 = 7 \cdot 2 + 0$

The last non-zero remainder is 7. So $\gcd(287, 91) = 7$.

GCDs as Linear Combinations

Theorem/Definition

Bézout's Theorem: If a, b are positive integers, then there exist integers s, t (called Bézout coefficients) such that $\gcd(a, b) = sa + tb$.

Question/Example

Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198. **Solution:** 1. Euclidean Algorithm:

- $252 = 1 \cdot 198 + 54$
- $198 = 3 \cdot 54 + 36$
- $54 = 1 \cdot 36 + 18$
- $36 = 2 \cdot 18 + 0$. So $\gcd = 18$.

2. Work backwards:

- $18 = 54 - 1 \cdot 36$ (from step 3)
- $18 = 54 - 1 \cdot (198 - 3 \cdot 54)$ (substitute 36 from step 2)
- $18 = 54 - 1 \cdot 198 + 3 \cdot 54 = 4 \cdot 54 - 1 \cdot 198$
- $18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198$ (substitute 54 from step 1)
- $18 = 4 \cdot 252 - 4 \cdot 198 - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$

So, $s = 4$ and $t = -5$. $\gcd(252, 198) = 4 \cdot 252 + (-5) \cdot 198$.

Theorem/Definition

Lemma 2: If $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$. **Lemma 3:** If p is prime and $p \mid (a_1 a_2 \dots a_n)$, then $p \mid a_i$ for some i .

Theorem/Definition

Theorem 7 (Dividing Congruences): If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$. (We can "cancel" c only if it's relatively prime to the modulus).

Section 4.4: Solving Congruences

Linear Congruences

Theorem/Definition

Definition: A **linear congruence** is of the form $ax \equiv b \pmod{m}$. **Definition:** An **inverse** of a modulo m is an integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$.

Theorem/Definition

Theorem 1: An inverse of a modulo m exists if and only if $\gcd(a, m) = 1$. The inverse is unique modulo m . **Proof Idea:** $\gcd(a, m) = 1 \implies sa + tm = 1$ (Bézout). Then $sa = 1 - tm \implies sa \equiv 1 \pmod{m}$. So s is an inverse.

Finding Inverses: Use the Extended Euclidean Algorithm (working backwards from finding the gcd, as shown in the previous section to find Bézout coefficients s, t). The coefficient s is the inverse $\bar{a}$. If s is negative, add m to get a positive inverse in Z_m .

Question/Example

Find an inverse of 3 modulo 7. **Solution:** $\gcd(3, 7) = 1$. Euclidean Alg: $7 = 2 \cdot 3 + 1$. Work backwards: $1 = 7 - 2 \cdot 3 = 1 \cdot 7 + (-2) \cdot 3$. Bézout coefficients are $s = -2, t = 1$. The inverse is $s = -2$. Since $-2 \equiv 5 \pmod{7}$, the unique inverse in Z_7 is 5. Check: $5 \cdot 3 = 15 \equiv 1 \pmod{7}$.

Question/Example

Find an inverse of 101 modulo 4620. **Solution:** 1. Euclidean Algorithm:

- $4620 = 45 \cdot 101 + 75$
- $101 = 1 \cdot 75 + 26$
- $75 = 2 \cdot 26 + 23$
- $26 = 1 \cdot 23 + 3$
- $23 = 7 \cdot 3 + 2$
- $3 = 1 \cdot 2 + 1$
- $2 = 2 \cdot 1 + 0$. $\gcd = 1$.

2. Work backwards:

- $1 = 3 - 1 \cdot 2$
- $1 = 3 - 1 \cdot (23 - 7 \cdot 3) = 8 \cdot 3 - 1 \cdot 23$
- $1 = 8 \cdot (26 - 1 \cdot 23) - 1 \cdot 23 = 8 \cdot 26 - 9 \cdot 23$
- $1 = 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75$
- $1 = 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75 = 26 \cdot 101 - 35 \cdot 75$
- $1 = 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101) = 26 \cdot 101 - 35 \cdot 4620 + 1575 \cdot 101$
- $1 = 1601 \cdot 101 - 35 \cdot 4620$

Bézout coefficients are $s = 1601, t = -35$. The inverse is $s = 1601$. Since $0 \leq 1601 < 4620$, this is the unique positive inverse.

Solving Linear Congruences: To solve $ax \equiv b \pmod{m}$ where $\gcd(a, m) = 1$: 1. Find the inverse $\bar{a}$ of a modulo m . 2. Multiply both sides by $\bar{a}$: $\bar{a}(ax) \equiv \bar{a}b \pmod{m}$. 3. This simplifies to $1 \cdot x \equiv \bar{a}b \pmod{m}$. 4. The solutions are all integers x such that $x \equiv \bar{a}b \pmod{m}$.

Question/Example

Solve $3x \equiv 4 \pmod{7}$. **Solution:** We found the inverse of 3 mod 7 is $\bar{a} = 5$. Multiply by 5: $5 \cdot (3x) \equiv 5 \cdot 4 \pmod{7}$. $15x \equiv 20 \pmod{7}$. Since $15 \equiv 1 \pmod{7}$ and $20 \equiv 6 \pmod{7}$, this becomes: $1 \cdot x \equiv 6 \pmod{7}$. So, $x \equiv 6 \pmod{7}$. Solutions are ..., -1, 6, 13, ...

The Chinese Remainder Theorem (CRT)

Used to solve systems of simultaneous congruences with pairwise relatively prime moduli.

Theorem/Definition

Theorem 2 (CRT): Let $m_1, \dots, m_n$ be pairwise relatively prime positive integers > 1 . The system: $x \equiv a_1 \pmod{m_1}$ $x \equiv a_2 \pmod{m_2}$... $x \equiv a_n \pmod{m_n}$ has a unique solution modulo $m = m_1 m_2 \dots m_n$.

Construction of Solution: 1. Let $m = m_1 m_2 \dots m_n$. 2. For each $k = 1, \dots, n$, let $M_k = m/m_k$. 3. For each k , find y_k , the inverse of M_k modulo m_k (i.e., $M_k y_k \equiv 1 \pmod{m_k}$). 4. The solution is $x = a_1 M_1 y_1 + a_2 M_2 y_2 + \dots + a_n M_n y_n$. The unique solution in Z_m is $x \pmod{m}$.

Question/Example

Solve Sun-Tsu's problem: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$. **Solution:** 1. $m_1 = 3, m_2 = 5, m_3 = 7$. They are pairwise relatively prime. $m = 3 \cdot 5 \cdot 7 = 105$. 2. $M_1 = 105/3 = 35$. $M_2 = 105/5 = 21$. $M_3 = 105/7 = 15$. 3. Find inverses: - $M_1 = 35 \equiv 2 \pmod{3}$. Inverse y_1 ? $2y_1 \equiv 1 \pmod{3}$. $y_1 = 2$ works ($4 \equiv 1 \pmod{3}$). - $M_2 = 21 \equiv 1 \pmod{5}$. Inverse y_2 ? $1y_2 \equiv 1 \pmod{5}$. $y_2 = 1$ works. - $M_3 = 15 \equiv 1 \pmod{7}$. Inverse y_3 ? $1y_3 \equiv 1 \pmod{7}$. $y_3 = 1$ works. 4. $x = a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3$ $x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1$ $x = 140 + 63 + 30 = 233$. 5. Unique solution mod 105: $x = 233 \pmod{105} = 23$. (Since $233 = 2 \cdot 105 + 23$). Check: $23 \equiv 2 \pmod{3}$, $23 \equiv 3 \pmod{5}$, $23 \equiv 2 \pmod{7}$. Correct.

Back Substitution Method: Alternative way to solve CRT problems.

Question/Example

Solve $x \equiv 1 \pmod{5}$, $x \equiv 2 \pmod{6}$, $x \equiv 3 \pmod{7}$ using back substitution. **Solution:** 1. From $x \equiv 1 \pmod{5}$, we know $x = 5t + 1$ for some integer t . 2. Substitute into second congruence: $5t + 1 \equiv 2 \pmod{6} \implies 5t \equiv 1 \pmod{6}$. Multiply by inverse of 5 mod 6. Since $5 \equiv -1 \pmod{6}$, the inverse is also $-1 \equiv 5 \pmod{6}$. $5(5t) \equiv 5(1) \pmod{6} \implies 25t \equiv 5 \pmod{6} \implies 1t \equiv 5 \pmod{6}$. So $t = 6u + 5$. 3. Substitute t back into x : $x = 5(6u + 5) + 1 = 30u + 25 + 1 = 30u + 26$. 4. Substitute into third congruence: $30u + 26 \equiv 3 \pmod{7}$. Since $30 \equiv 2 \pmod{7}$ and $26 \equiv 5 \pmod{7}$, this is $2u + 5 \equiv 3 \pmod{7}$. $2u \equiv 3 - 5 \equiv -2 \equiv 5 \pmod{7}$. Multiply by inverse of 2 mod 7. $2 \cdot 4 = 8 \equiv 1 \pmod{7}$, so inverse is 4. $4(2u) \equiv 4(5) \pmod{7} \implies 8u \equiv 20 \pmod{7} \implies 1u \equiv 6 \pmod{7}$. So $u = 7v + 6$. 5. Substitute u back into x : $x = 30(7v + 6) + 26 = 210v + 180 + 26 = 210v + 206$. 6. Final solution: $x \equiv 206 \pmod{210}$. Check: $206 \equiv 1 \pmod{5}$. $206 = 34 \cdot 6 + 2 \implies 206 \equiv 2 \pmod{6}$. $206 = 29 \cdot 7 + 3 \implies 206 \equiv 3 \pmod{7}$. Correct.

Fermat's Little Theorem

Theorem/Definition

Theorem 3 (Fermat's Little Theorem): If p is prime and a is an integer not divisible by p (i.e., $\gcd(a, p) = 1$), then: $a^{p-1} \equiv 1 \pmod{p}$. Furthermore, for any integer a (including when $p \mid a$): $a^p \equiv a \pmod{p}$.

Question/Example

Find $7^{222} \pmod{11}$. **Solution:** $p = 11$ is prime, $a = 7$ is not divisible by 11. By Fermat's Little Theorem, $7^{11-1} = 7^{10} \equiv 1 \pmod{11}$. Now, $222 = 10 \cdot 22 + 2$. So, $7^{222} = 7^{10 \cdot 22 + 2} = (7^{10})^{22} \cdot 7^2$. $7^{222} \equiv (1)^{22} \cdot 7^2 \pmod{11}$ $7^{222} \equiv 1 \cdot 49 \pmod{11}$ $7^{222} \equiv 49 \pmod{11}$ Since $49 = 4 \cdot 11 + 5$, we have $49 \equiv 5 \pmod{11}$. Therefore, $7^{222} \pmod{11} = 5$.

Pseudoprimes

Theorem/Definition

Definition (Pseudoprime): A composite integer n is a pseudoprime to base b if $b^{n-1} \equiv 1 \pmod{n}$. (If $b^{n-1} \not\equiv 1 \pmod{n}$, then n must be composite. If $b^{n-1} \equiv 1 \pmod{n}$, n could be prime or pseudoprime). **Definition (Carmichael Number):** A composite integer n which is a pseudoprime to *every* base b with $\gcd(b, n) = 1$. Example: $561 = 3 \cdot 11 \cdot 17$.

Question/Example

341 is a pseudoprime to base 2. $341 = 11 \cdot 31$ (composite). Check if $2^{340} \equiv 1 \pmod{341}$. (Verification involves modular exponentiation, mentioned in slides).

Primitive Roots and Discrete Logarithms

Theorem/Definition

Definition (Primitive Root): An integer r is a primitive root modulo prime p if every integer a with $1 \leq a \leq p-1$ is congruent to a power of r modulo p . (i.e., the powers $r^1, r^2, \dots, r^{p-1}$ generate all non-zero elements of $\mathbb{Z}_p$). **Fact:** Primitive roots exist for every prime p .

Definition (Discrete Logarithm): If p is prime, r is a primitive root mod p , and $a \equiv r^e \pmod{p}$ with $1 \leq e \leq p-1$, then e is the discrete logarithm of a modulo p to the base r . Notation: $\log_r a = e$.

Question/Example

Modulo 11:

- Powers of 2: $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 5, 2^5 = 10, 2^6 = 9, 2^7 = 7, 2^8 = 3, 2^9 = 6, 2^{10} = 1$. Generates all numbers from 1 to 10. So, 2 is a primitive root modulo 11.
- Powers of 3: $3^1 = 3, 3^2 = 9, 3^3 = 5, 3^4 = 4, 3^5 = 1$. Only generates $\{1, 3, 4, 5, 9\}$. So, 3 is NOT a primitive root modulo 11.
- $\log_2 3 = 8 \pmod{11}$ since $2^8 \equiv 3 \pmod{11}$.
- $\log_2 5 = 4 \pmod{11}$ since $2^4 \equiv 5 \pmod{11}$.

Note: Computing discrete logarithms is generally considered computationally hard for large primes, which is important for cryptography.